FAIRNESS AUDIT · RANDOM FOREST & GRADIENT BOOSTING · CDC BRFSS 2022

Do heart-attack prediction models perform equally across sex, race and age?

Two tree-ensemble classifiers were trained on the Personal Key Indicators of Heart Disease survey and evaluated separately for each demographic group. Group scores were compared with independent-samples t-tests over 5 test batches and with bootstrap 95% confidence intervals on the full test group.

246,022	5.46%	RF · GB	3
complete survey records	heart-attack prevalence	models audited	attributes: sex, race, age

Performance gaps at a glance

Difference in score between groups (first group minus second). Filled cells are significant at $p < 0.05$ by bootstrap (group difference, 95% CI).

	Random Forest				Gradient Boosting			
COMPARISON	ACCU.	PREC.	RECA.	F1	ACCU.	PREC.	RECA.	F1
Sex Male − Female	+0.001	**+0.054**	**+0.119**	**+0.086**	**−0.027**	+0.018	**+0.213**	**+0.231**

Race Black − White	−0.036	**−0.079**	−0.023	**−0.052**	−0.033	**−0.071**	−0.035	**−0.053**
Age Under 50 − 50+	**+0.125**	+0.033	**−0.371**	**−0.222**	**+0.125**	+0.045	**−0.360**	**−0.218**

first group scores higher second group scores higher not significant

Group comparison explorer

Pick a model and a demographic attribute.

Random Forest | Gradient Boosting

Sex | Race | Age

Male

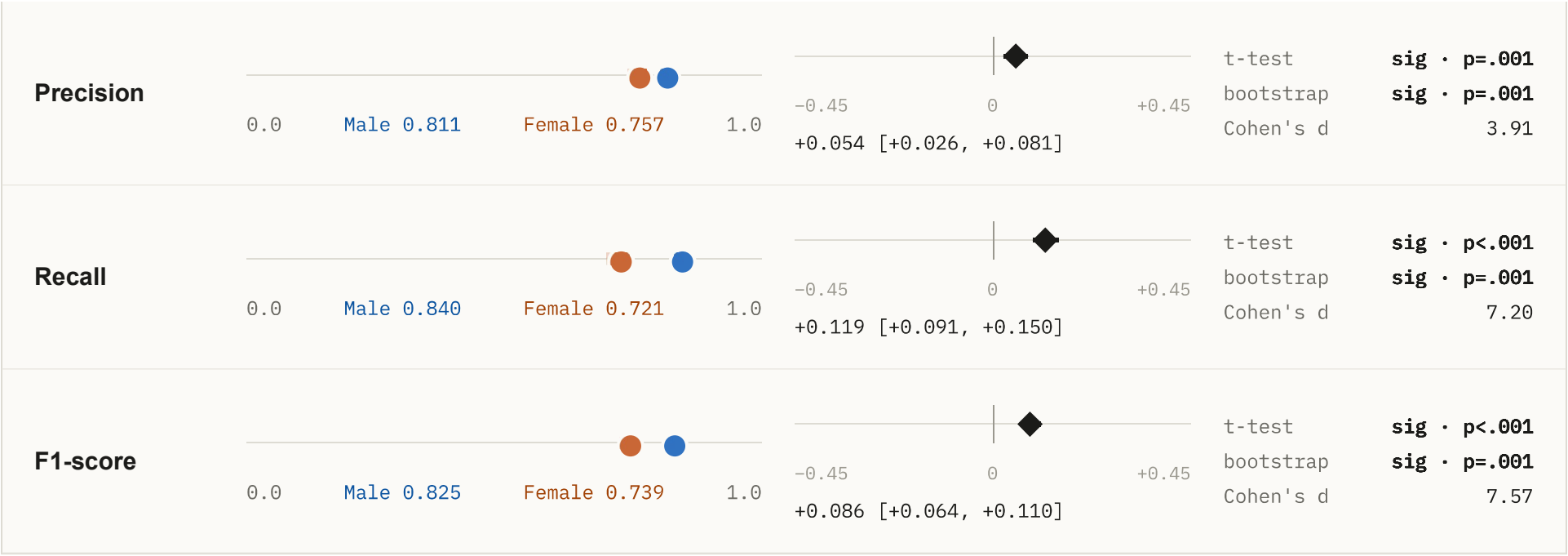
4,103 test cases 2,373 heart attacks 57.8% prevalence in test group

Female

3,397 test cases 1,377 heart attacks 40.5% prevalence in test group

> Accuracy is identical for men and women (0.794 vs 0.793), but the model finds 84% of male heart attacks and 72% of female ones. Precision and F1 are also significantly lower for women. Reporting accuracy alone would hide this gap.

METRIC	GROUP MEAN (5 BATCHES) ± SD	DIFFERENCE · BOOTSTRAP 95% CI	TESTS
Accuracy	0.0 Male 0.794 Female 0.793 1.0	−0.45 0 +0.45 +0.001 [−0.018, +0.018]	t-test p=.907 bootstrap p=.924 Cohen's d 0.08



Test cases come from the balanced 25,000-record sample (7,500 test cases, 50% heart attacks). Group prevalence differs from the population because of this balancing.

Overall model performance

All groups combined. The two models were trained on different data, so their overall scores are not directly comparable.

Random Forest

Balanced 25,000-record sample · held-out test set of 7,500

ACCURACY	PRECISION	RECALL	F1
0.793	0.792	0.796	0.794
CV 0.799	CV 0.801	CV 0.795	CV 0.798

Test scores closely match the 5-fold cross-validation means (std ≤ 0.013), so the model is stable across folds.

Gradient Boosting

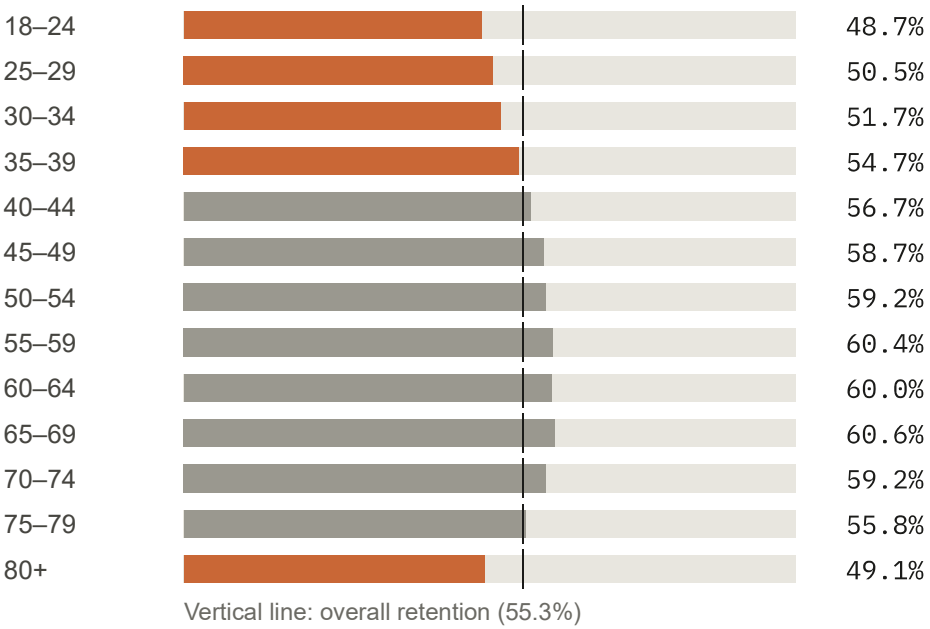
Full imbalanced data · held-out test set of 73,807

ACCURACY	PRECISION	RECALL	F1
0.949	0.568	0.241	0.339
CV 0.949	CV 0.580	CV 0.247	CV 0.346

Accuracy of 0.949 is close to the 94.5% achievable by always predicting "no heart attack". Recall of 0.241 means about three in four heart attacks are missed.

Records kept after dropping missing values

55.3% of the 445,132 original rows had no missing values. Retention was lowest at both ends of the age range, so the youngest and oldest respondents are under-represented in training.



Vertical line: overall retention (55.3%)

Method

Data	Personal Key Indicators of Heart Disease (Kaggle, 2022 BRFSS), complete-case file: 246,022 rows, 40 columns. Target: HadHeartAttack.
Random Forest	Balanced sample of 25,000 (17,500 train / 7,500 test, 57 features). Grid search over 18 combinations with 3-fold CV; best: 400 trees, gini, min_samples_split 5.
Gradient Boosting	400 estimators, learning rate 0.1, max depth 3. Sex audit on the full data (172,215 train / 73,807 test, 68 features); race and age audits on the same balanced 25,000 sample as RF.
Testing	Each group's test cases were split into 5 batches for independent-samples t-tests (with Cohen's d). A bootstrap on all group test cases gave 95% CIs and p-values for the difference.

Data: Kaggle · kamilpytlak/personal-key-indicators-of-heart-disease (2022)